

```
print("Training images shape:", x_train.shape)
print("Training labels shape:", y_train.shape)
```

```
print("Testing images shape:", x_test.shape)
print("Testing labels shape:", y_test.shape)
```

Downloading data from <https://storage.googleapis.com/11490434/11490434>  0s 0us/step

Training images shape: (60000, 28, 28)

Training labels shape: (60000,)

Testing images shape: (10000, 28, 28)

Testing labels shape: (10000,)

```
print("Number of training images:", len(X_train))
print("Number of testing images:", len(X_test))

print("Image size:", X_train[0].shape)

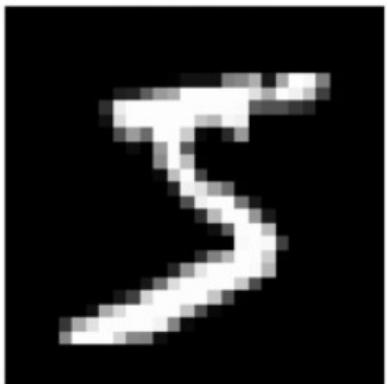
print("First image label:", y_train[0])

print("Minimum pixel value:", X_train.min())
print("Maximum pixel value:", X_train.max())
```

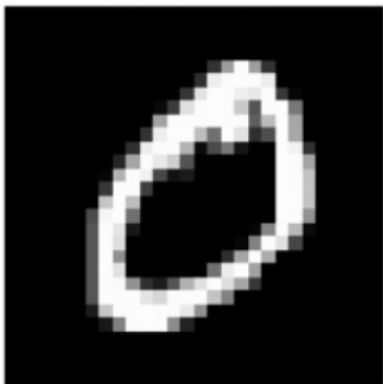
```
Number of training images: 60000
Number of testing images: 10000
Image size: (28, 28)
First image label: 5
Minimum pixel value: 0
Maximum pixel value: 255
```

plt.show()

Label: 5



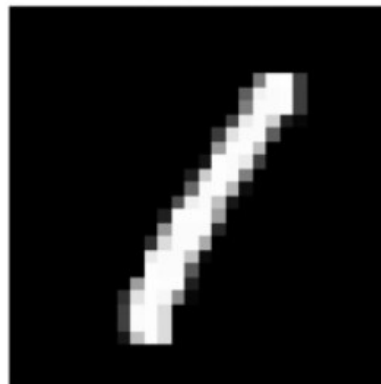
Label: 0



Label: 4



Label: 1



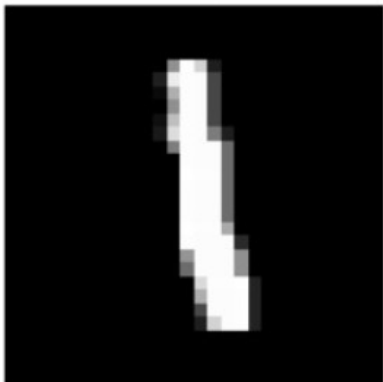
Label: 9



Label: 2



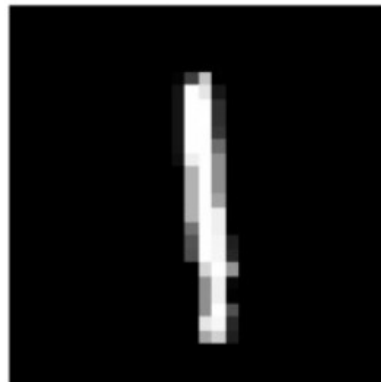
Label: 1



Label: 3



Label: 1



Label: 4



```
test_loss, test_accuracy = model.evaluate(X_test, y_test, verbose=0)

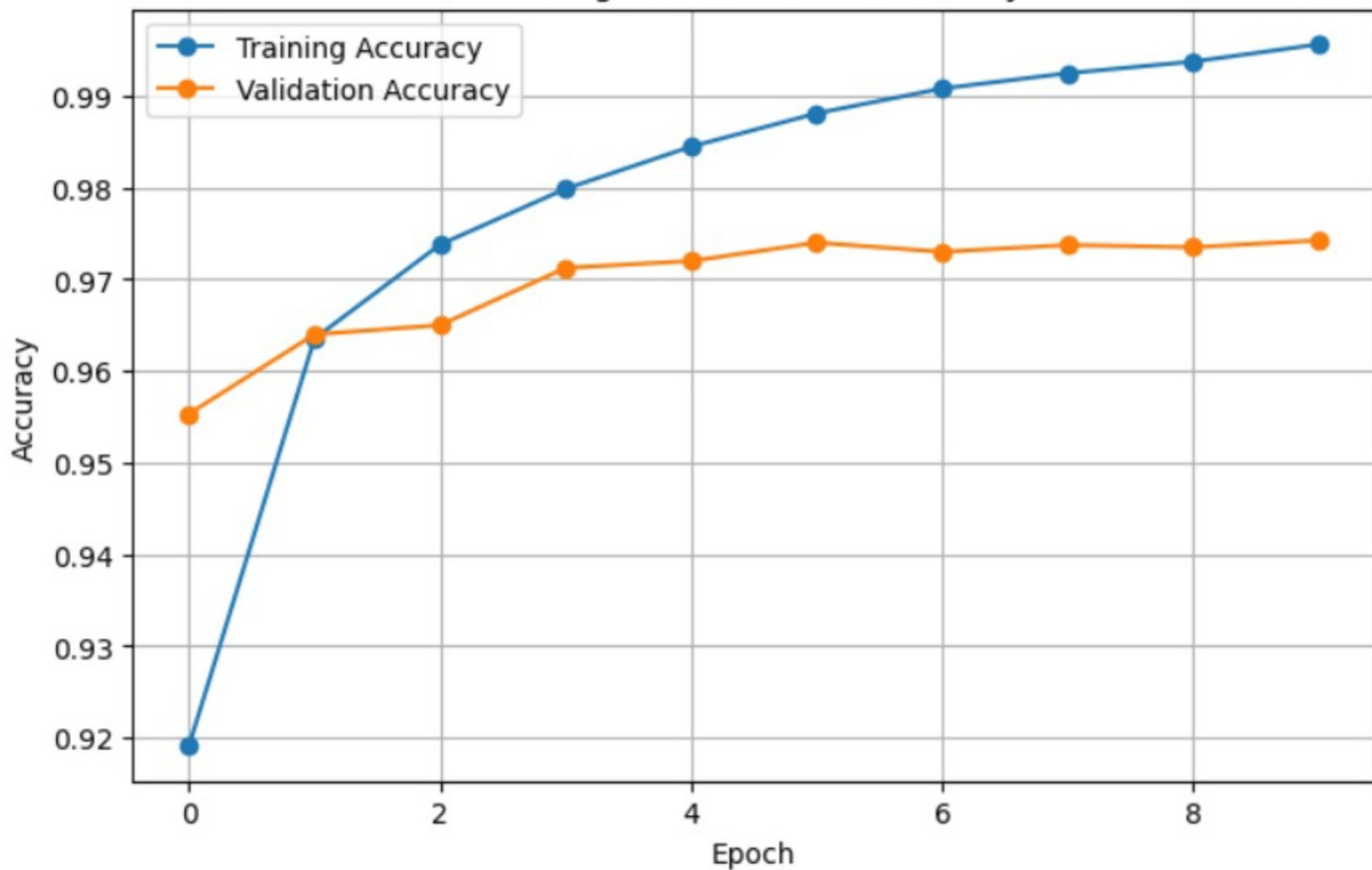
print("Test Loss:", test_loss)
print("Test Accuracy:", test_accuracy)
print("Test Accuracy Percentage:", test_accuracy * 100, "%")
```

Test Loss: 0.0889134481549263

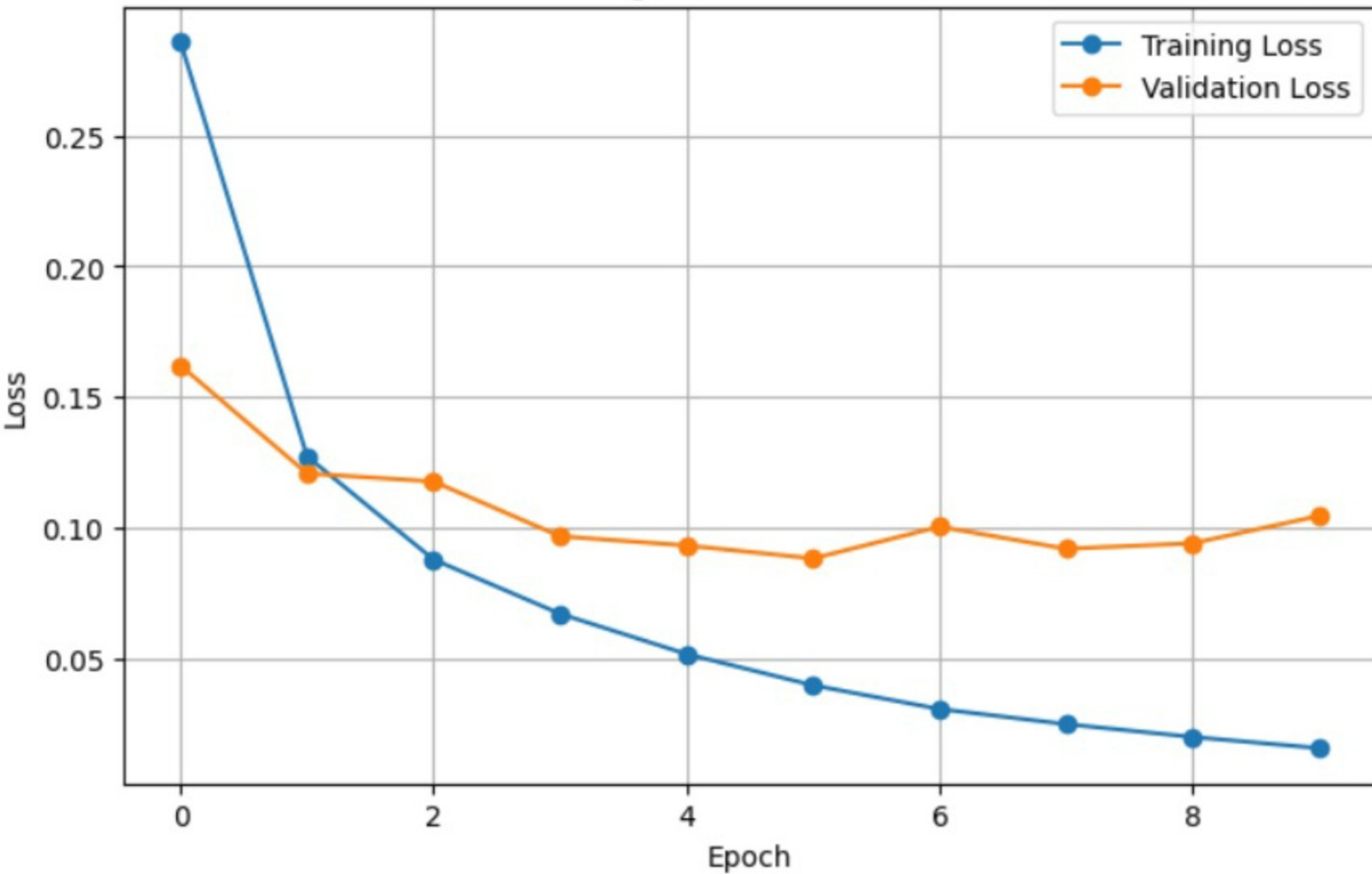
Test Accuracy: 0.9761000275611877

Test Accuracy Percentage: 97.61000275611877 %

Training and Validation Accuracy



Training and Validation Loss



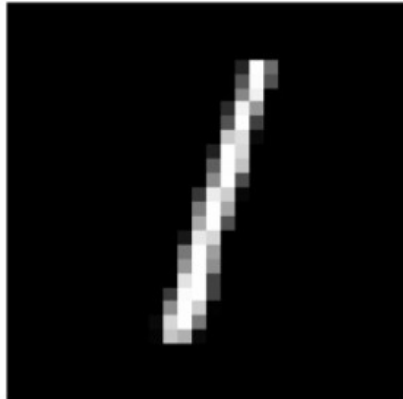
Actual: 7
Predicted: 7



Actual: 2
Predicted: 2



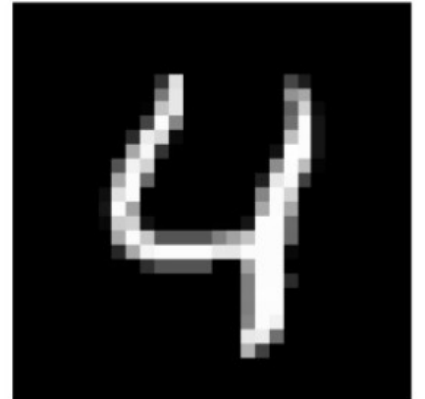
Actual: 1
Predicted: 1



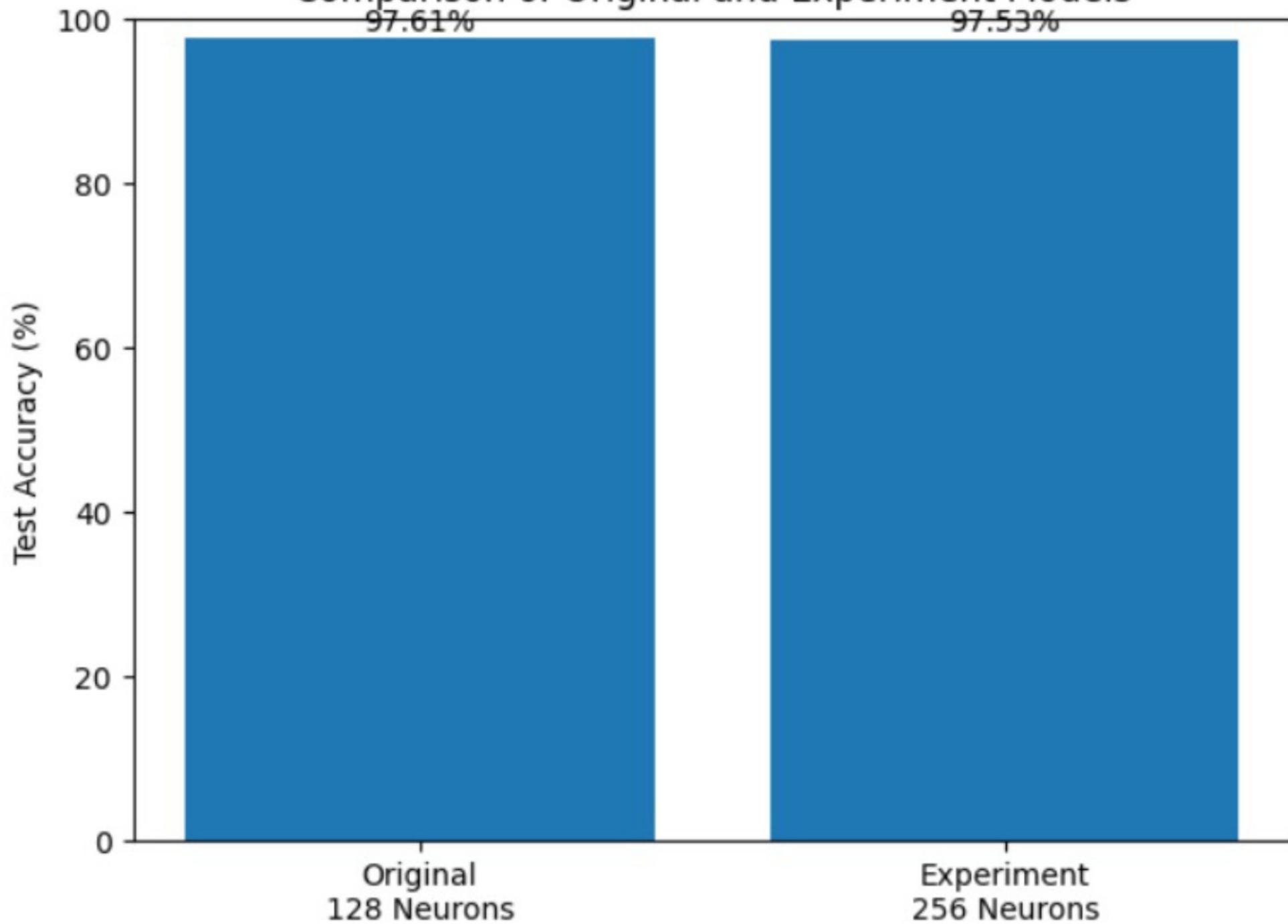
Actual: 0
Predicted: 0

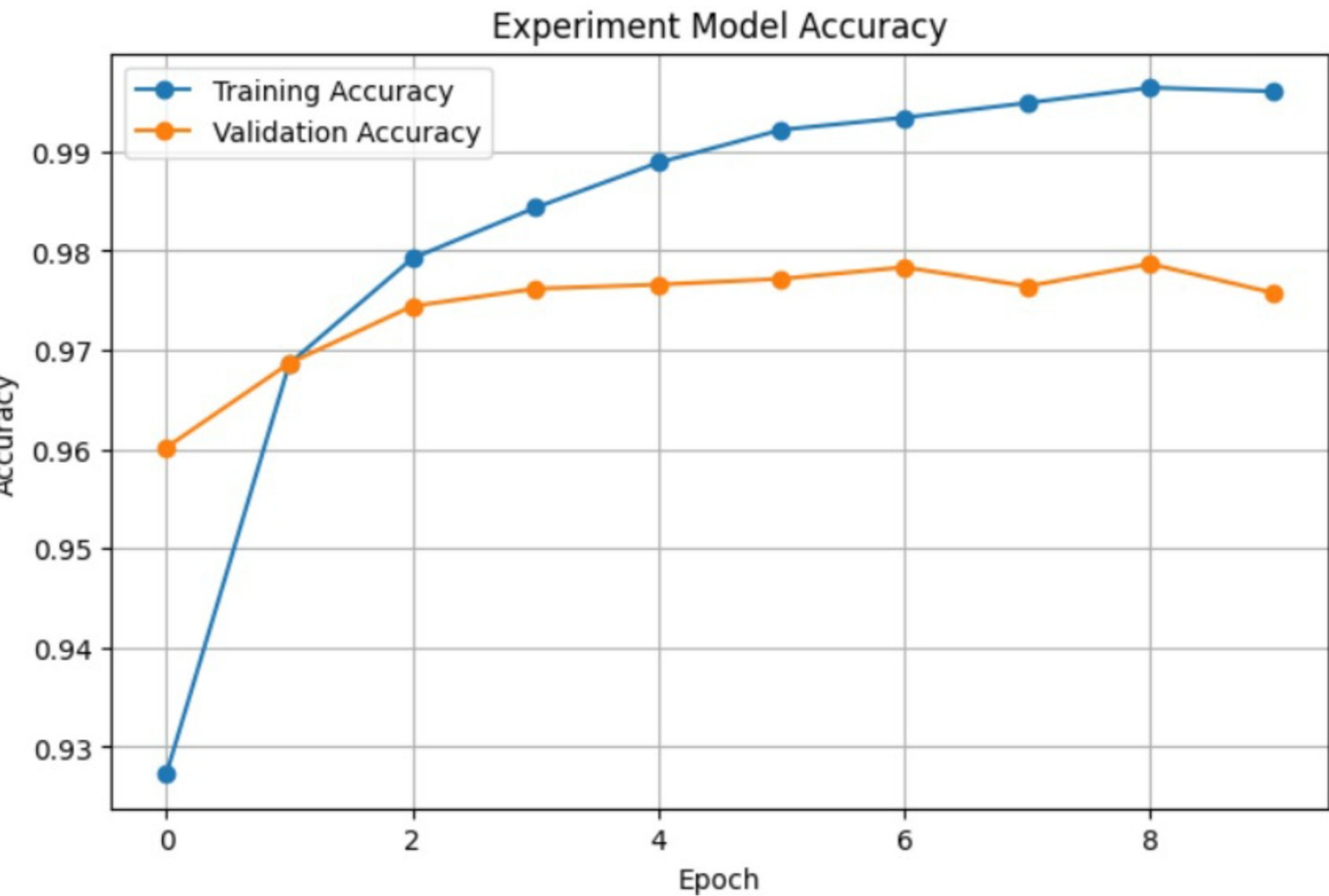


Actual: 4
Predicted: 4



Comparison of Original and Experiment Models





```
        f"Predicted = {predicted_labels[i]}"  
    )
```

FINAL RESULTS

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Original Model Test Accuracy: 97.61%

Experiment Model Test Accuracy: 97.53%

5 Image Predictions:

Image 1: Actual = 7, Predicted = 7

Image 2: Actual = 2, Predicted = 2

Image 3: Actual = 1, Predicted = 1

Image 4: Actual = 0, Predicted = 0

Image 5: Actual = 4, Predicted = 4